ON THE CALCULATION OF THE FREEZING POINT OF ICE-CREAM MIXES AND OF THE QUANTITIES OF ICE SEPARATED DURING THE FREEZING PROCESS*

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In studying the physics of ice-cream freezing and in calculating refrigeration constants for ice-cream work, it is desirable to be able not only to calculate the freezing points of ice-cream mixes but also to determine the quantity of ice that may be separated from the mix at any temperature. Making certain allowable assumptions, it is possible to calculate these values with considerable accuracy.

Van Slyke (1) gives the composition of the average milk as follows:

		per cent
Fat		3.9
Milk solids		8.99
Lactose	4.9	
Protein	3.2	
Salts	0.89	
Water		87.11

Because milk is an animal secretion it is of necessity isotonic with the blood of the animal from which it comes. This means that the freezing point of milk is a fairly constant value, which for cow's milk may be taken as −0.55°C., or in other words, the freezing point of milk is 0.55°C. lower than that of water.

By employing the usual formula for obtaining the molecular weight of an un-ionized substance from the depression of the freezing point of a solvent it can be calculated that the milk sugar accounts for 0.306°C. of the normal freezing-point lowering of milk and that 0.244°C. is caused by the combined action of the milk salts, protein, fat, etc. Not taking into consideration the

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very small effect of the fat and protein upon the freezing point of milk, it is found from these data that the *apparent* molecular weight of the milk salts is 78.6, a figure which will be of considerable value in the calculations given later in this paper.

The molecular weight formula is as follows:

$$\Delta = K\frac{G}{M}$$

In this formula Δ represents the freezing point depression in degrees Centigrade; K the constant, depending on the molecular weight of the solvent (for water 18.6); G the weight of dissolved substance in 100 grams of water; and M its molecular weight.

From the analysis of milk previously given, it is seen that there are 4.9 parts of lactose to 87.11 parts of water, which is equivalent to 5.63 parts of lactose to 100 parts of water. The substitution of the values in the molecular weight formula gives the following:

$$\Delta = 18.6\,\frac{5.63\,*}{342.2}$$

$$\Delta = 0.306$$

* For the purposes of this paper Van Slyke is considered to have expressed the lactose content of milk in terms of anhydrous lactose rather than in those of the monohydrate.

Subtracting 0.306 from 0.55 gives 0.244, the depression of the freezing point of water due to the salts of milk. By again using the molecular weight formula, the apparent molecular weight of the salts can be calculated by substituting 0.244 for Δ and, since there are 0.899 part salts to 87.101 parts water, or 1.032 parts salts to 100 parts water, by substituting 1.032 for G:

$$M = K\frac{G}{\Delta}$$

$$M = 18.6\,\frac{1.032}{0.244}$$

$$M = 78.6$$

From the molecular weight formula and from a knowledge of the composition of an ice-cream mix, the depression of the freezing point of the mix that is caused by the salts can be calculated.

TABLE 1

Freezing point lowering of cane sugar solutions

PARTS CANE SUGAR TO 100 PARTS WATER	PER CENT CANE SUGAR	LOWERING	LOWERING DUE TO 1 PART CANE SUGAR
		°C.	
3.59	3.47	0.21	0.05
6.85	6.41	0.40	0.05
10.84	9.78	0.65	0.06
15 83	13 67	0 95	0.06
19.80	16.53	1.23	0.06
22.58	18.42	1.37	0.06
25.64	20.41	1.58	0.06
28.51	22.19	1.77	0.06
32.22	24.37	1.99	0.06
35.14	26.00	2.15	0.06
37.86	27.46	2.33	0.06
43.72	30.42	2.71	0.06
45.62	31.33	2.82	0.07
50.02	33.35	3.13	0.07
54.74	35.37	3.47	0.07
59.46	37.29	3.81	0.07
64.55	39.23	4.22	0.07
69.74	41.09	4.60	0.07
75.91	43.15	5.07	0.07
82.35	45.16	5.65	0.07
88.67	47.00	6.11	0.07
95.94	48.97	6.76	0.07
102.70	50.65	7.38	0.07
111.30	52.67	8.06	0.07
121.00	54.75	9.02	0.07
131.60	56.82	9.93	0.07
143.10	58.86	10.90	0.07
153.80	60.60	11.69	0.08
165.60	62.35	12.72	0.08
181.70	64.49	13.80	0.08

The other ingredients of the mix that will affect the freezing point are the two sugars, sucrose and lactose. By using the method previously given, it would be possible to calculate the freezing-point depression of an ice-cream mix caused by these

sugars if it had not been shown (2) that sucrose in concentration does not obey the freezing-point law. To find the freezing-point depression then, it is necessary to calculate the *total* sugar concentration of an ice-cream mix, on the water basis, and refer to the actual freezing-point curve for cane-sugar solutions, which has been worked out quite carefully by Pickering (3) and

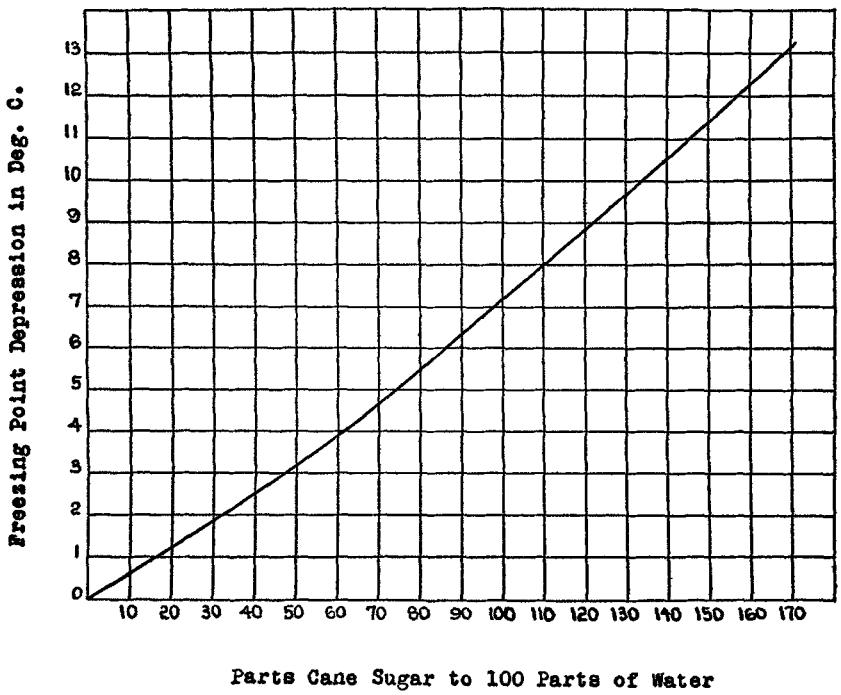

FIG. 1. FREEZING-POINT DEPRESSIONS OF CANE SUGAR SOLUTIONS

checked by P. N. Peter (4) of these laboratories. By adding to this freezing-point depression the depression caused by the salts, a very close approximation of the true freezing point of the mix is obtained.

Two assumptions have been made here which have not yet been proved experimentally: First, that the freezing-point depression caused by the salts obeys the freezing-point law, which is probably true because of the moderately concentrated solutions

encountered; second, that mixtures of lactose and cane sugar, where lactose is present in moderate amounts, will closely follow the freezing-point curve of pure cane sugar in water solution, because the molecular weight of the lactose (anhydrous) is the same as that of cane sugar.

The simplest way of testing the validity of these assumptions is to calculate and determine experimentally the freezing points of some actual ice-cream mixes. For this calculation the preceding method can be simplified somewhat.

Multiplying the number parts milk solids not fat in the mix by 0.545 gives the lactose content. Adding to this the cane-sugar content, multiplying by 100 and dividing by the number parts water give the number parts sugar per 100 parts water. This may be expressed as follows:

$$\frac{(\text{Milk solids not fat} \times 0.545 + \text{Sucrose})\ 100}{\text{Water}} = \text{Parts sugar to 100 of water}$$

Referring then to the freezing-point curve of sugar, the corresponding freezing-point depression (A) is found.

Dividing the number parts milk solids by ten gives the salts content of the mix. Multiplying by 100 and dividing by the water content give the number parts salts to 100 parts water, and this multiplied by 18.6 and divided by 78.6 give the freezing-point depression of the salts. This may also be expressed as follows:

$$\frac{\text{M.S.N.F.} \times 100 \times 18.6}{10 \times \text{Water} \times 78.6} = \text{Freezing point depression due to salts } (B)$$

or

$$\frac{\text{M.S.N.F.} \times 2.37}{\text{Water}} = B$$

$$A + B = \text{Total Lowering}$$

These formulas are now applied to the calculation of the freezing points of several ice-cream mixes made up in such a way that the only difference between them is in the water concentration. This is done in order to calculate and check the

freezing-point curve of a certain basic mix which had the following concentration:

		parts
Total solids		36.03
Fat	12.5	
Sugar (sucrose)	14.0	
Milk solids not fat	9.53	
Water		63.97

By concentration in the vacuum pan, samples of this mix were made which had one-eighth, one-quarter, three-eighths, and one-half of the water removed. The freezing points of these concentrations were determined with the Beckman thermometer in a standard freezing-point apparatus and define the freezing

TABLE 2

Results of calculations and measurements of freezing points of experimental ice-cream mixes

NUMBER	FAT	M.S.N.F.	CANE SUGAR	WATER	TOTAL SUGAR TO 100 H_2O	F.P. DEP. DUE TO SUGAR A	F.P. DEP. DUE TO SALTS B	CAL. F.P. $\left(\begin{matrix}A\\+\\B\end{matrix}\right)$	F.P.
	parts	*parts*	*parts*	*parts*	*parts*				
1	12.5	9.53	14.00	63.97	30.00	1.90	0.35	−2.25	−2.29
2	12.5	9.53	14.00	55.98	34.29	2.12	0.40	−2.52	−2.54
3	12.5	9.53	14.00	47.98	40.00	2.47	0.47	−2.94	−3.02
4	12.5	9.53	14.00	40.00	47.86	2.98	0.56	−3.54	−3.62
5	12.5	9.53	14.00	31.99	60.00	3.83	0.71	−4.54	−4.49

point curve of the basic mix. Table 2 gives the results of these measurements as well as of the calculations made by the formulas previously outlined.

It will be seen that there is a very fair agreement between the calculated values and those obtained experimentally.

Experiments appear to justify fully the assumption that the temperature of freezing of an ice-cream mix concentrated in the vacuum pan is the same as the equilibrium temperature that must be attained when the unfrozen portion of the partially frozen unconcentrated mix has the same concentration as the vacuum-prepared sample. These experiments seem to show that ice-cream mixes can be moderately concentrated in the pan

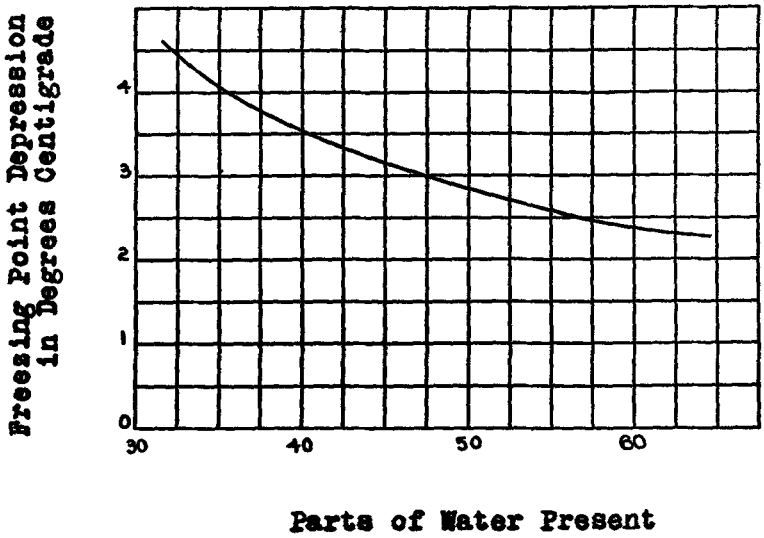

FIG. 2. FREEZING-POINT DEPRESSIONS OF MIXES PLOTTED AGAINST WATER CONCENTRATION

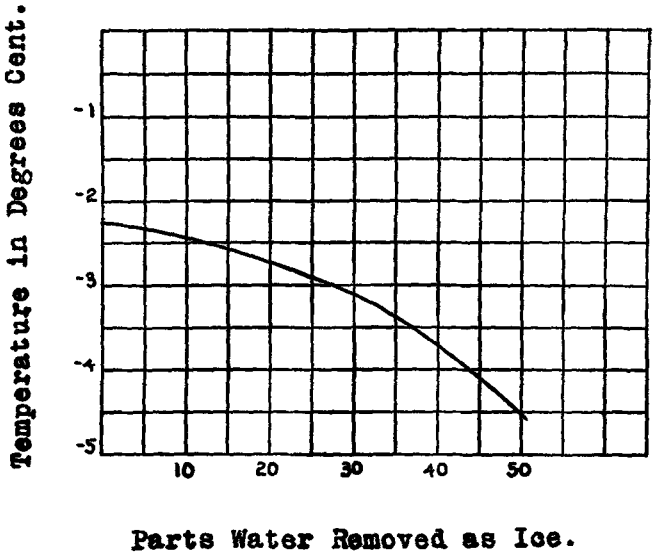

FIG. 3. QUANTITIES OF ICE SEPARATED FROM EXPERIMENTAL MIX AT DIFFERENT TEMPERATURES

and rediluted to normal concentration without any apparent effect upon the physical properties of the mix.

In figure 2 is plotted the freezing point depressions of the experimental mixes against the water concentrations. In figure 3 the freezing points are plotted against the portions of water removed from the basic mix. This latter curve can be used to determine the quantity of ice that may be separated from the ice-cream mix at any freezing temperature. The best way therefore, to determine the quantity of ice that may be separated from any mix at any temperature is to calculate the freezing points of the mix at a number of concentrations, then to plot the freezing-point water-removed curve and from this to estimate the quantity of ice that is frozen out at any one temperature.

Two questions would now naturally arise. First, with the supercooling of the mix in the freezer, how can the equilibrium temperature be determined? Second, may milk sugar crystallizing in the freezer invalidate our calculations, since these calculations are really based upon the assumption that ice is the only material to be thrown out in solid form during the period that the mix is in the freezer?

Unpublished experiments carried out by O. E. Williams and the author show that if a small portion of ice cream is removed from the freezer and its temperature determined, this sample is no longer supercooled and the temperature is very close to the equilibrium temperature. This is true because the quantity of ice separating to attain equilibrium after the portion is removed from the freezer is negligible.

In answer to the second question it can be said that the tendency of lactose to crystallize in the freezer will probably be very slight. This conclusion is reached from the following consideration. Hudson (5) has shown that when lactose is in equilibrium in solution, one part is in the hydrated form and one and one-half parts in the anhydrous form. The anhydrous form is very soluble. When lactose crystallizes, the hydrate deposits, some of the anhydride turns to hydrate and also deposits, the rate of separation then being governed by the rate of transformation, which at low temperatures is shown to be very slow.

The most concentrated mix considered here has about sixteen parts of lactose to 100 parts of water. At −4.5°C. probably about nine parts are soluble. This would mean that our solution is supersaturated to at least seven parts. However, since at least four of these are in the anhydrous form, this solution, while it is potentially supersaturated to at least seven parts, is actually supersaturated to but three. In view of the great tendency (6) of lactose to supersaturate, it seems unlikely that any should crystallize in the freezer. Even if this should happen, calculation shows that the resultant error in determining the quantity of ice that would be frozen at this temperature (−4.5°C.) is about one per cent. At higher temperatures this error would be proportionately less.

In giving the method for calculating the freezing points of ice-cream mixes and in showing how the quantity of ice that may be separated from a mix at any temperature in the freezer can be determined, no reference has been made to the effect of flavoring material on the freezing point of the basic mixes. However, if the water concentration, freezing point, and quantity of flavoring material added to the mix are known, the effect on the freezing points of the basic mixes may be calculated by the principles outlined in this paper. Allowance must be made if any milk product used in making the mix is low in lactose content.

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